Quality Control Protocol - Zeiss Confocal Microscopes

Objectives:

5x Fluar, NA = 0.25, Air

10x N-Achroplan, NA = 0.3, Water

20x Plan-Apochromat, NA = 1.0, Water

25x Plan-Apochromat, NA = 0.8, Water, Silicone Oil, Glycerol and Oil

Necessary equipment:

- Zeiss calibration objective (5x)
- Printed copy of laser power template for the microscope
- Thorlabs laser power meter PM100D
- Thorlabs power meter slide-shaped sensor S170C
- Chroma slides
- Ceramic dots slide
- Fluorescent beads: 170nm green (505nm emission), 500nm multicolor (430+515+580+680 emission)
    - Check that beads are not older than one year
    - For protocol, see: J:\Methods and Protocols\Sample preparation\ FluorescentBeads_updated250324.docx
- For cleaning:
    - Zeiss lens cleaning solution (Isopropanol + n-Hexane)
        - For recipe, see: J:\Methods and Protocols\Cleaning and maintaining optics\ The_Clean_Microscope_Zeiss.pdf
    - Dust blower
    - Cotton-tipped wooden sticks
    - Distilled water

<u>Start-up:</u>

- Create a sub-folder with current date in

  J:\Equipment\Microscopes\Imaging Facility\[microscope]\QualityControl sub-folder name: yymmdd

- Switch system on, using the switches on equipment “in the box”, not on top.
- Ensure that laser key is “on”
- Let the instrument warm up 60 min before start of QC measurements
- Clean all objectives and oculars:

    - Use the dust blower to remove loose particles from front and back lenses
    - Use distilled water and a cotton swab to remove water-soluble dirt
    - Use Zeiss Brillenreiniger and a cotton swab to remove oily dirt by moving swab in an outwards spiral
    - Inspect lens; if it is not yet clean, use Zeiss cleaning solution and a cotton swab. Again, use an outward spiral motion, from the center to the edge of the lens.
- Check all objectives for signs of damage:
    - Use inverted ocular to inspect the front lens and back lens
- Wipe dust and any spills from microscope and surrounding area
- Check that there's enough (and refill if needed):
    - immersion medium
    - EtOH
    - glasses cleaner
    - lens wipes
    - Kim wipes
- Note temperature in the room if possible:

Checks using System Maintenance and Calibration tool:

- Mount the Zeiss calibration objective in **Position 1**: enter ID number via touch screen.
- Go to Tools → System maintenance and calibration
- Select User group → Maintenance
- Enter the password: "service"
- **Scanfield test and scanner calibration:**
    - Run the scanfield test (approx. 20 min.)
    - Restart ZEN software
    - Run scanner calibration (approx. 13 min.)
    - Restart ZEN software
    - Run scanfield test again
- **Run Laser power linearization** (approx. 2 min.)
    - **If this option is not available in the software, do NOT run Laser power optimization instead!**
    - Laser power optimization should only be run after advice from a Zeiss service engineer to do so.
- Remove calibration objective and replace the objective there previously.

Check Z flatness of stage:

Note: This segment may be completed before the lasers have fully warmed up; i.e., within the first hour of turning on the system.

Sample: ceramic dot slide

Objective: 5x air, low NA

Procedure:

- Find the focal plane using epifluorescence.
- Mark four dots using settings for Alexa488 or similar, one in each corner of the slide.
    - Name them topleft, bottomright, etc., to indicate location

- Set a z-stack that encompasses at least the majority of all four dots
- Image a stack of each dot
- If the stacks are not centered, adjust stage using hex key to lower or raise one side
- Repeat imaging and adjusting until stacks align with each other
- Note down which z-position corresponds to the focal plan for each dot. Calculate the difference in height using the z-step information.
    - Example:
    - topleft is in focus at z-plane 3
    - topright is in focus at z-plane 6
    - the z-step size is 6.5µm
    - (6-3)(6.5) = 19.5 = the difference in focal plane from left to right.
- Note this data in the QC Results Summary sheet.
- Saving data is not necessary for this segment.

<u>Laser Power:</u>

Sample: ThorLabs power meter slide S170C

Deactivate laser blanking

Objective: 5x air, low NA

Zoom: 40x

Bidirectional scanning

Detector "on"

"High intensity laser range" checked

Procedure:

- Mount Thorlabs Laser Power Meter sensor S170C
    - Do not focus the laser on the sensor, because focus does not influence power measurement, but may damage the sensor
- Use highest possible zoom
- Deactivate laser blanking by going to system maintenance and calibration → extras → Blanking switch and more → "Disable blanking"
- Switch off room light
- For each laser, wavelength, and percentage on the template sheet:
    - Set the power meter to the correct wavelength
    - On the power meter, press Δ twice to set the reference value
    - Adjust the laser's percentage
    - Start scanning
    - Record the power meter value in µW next to that wavelength/percentage on the sheet
- Return to Zoom 1
- Re-enable blanking

<u>Detector Dark Noise:</u>

Sample: None

Objective: 25x

Pixels: 512 x 512

Zoom: 1

Speed: 6

Averaging: None

Procedure:

- Switch to 25x objective
- Close the pinhole as much as possible
- Open Detector Darknoise template from local QC folder
    - Acquire one multichannel image, with one channel per detector
- Shut off room light
- Make the acquisition by pressing “Snap”
- Save project from today as “detector_darknoise”
- Put pinhole back to normal size; 1 AU

<u>Detector Gain Linearity:</u>

Sample: yellow-green Chroma slide

Objective: 5x air, low NA

Pixels: 512 x 512

Speed: 6

Averaging: 4 lines

Procedure:

- Open Detector_Gain_Linearity_Template.czexp from LSM900 QC folder to use as template
- Set save path to QC folder on J: drive
- Acquire an image with each of these gains, for each detector:
    - 400, 500, 600, 700, 800, 900, 1000V
- Adjust laser power so that image is bright, but not saturated, at the highest gain setting. The laser power should be the same for all detectors. \
- Press “start experiment”
- Save project from today as “detector_gain_linearity”
- Close both projects

<u>Field illumination homogeneity:</u>

Sample: Yellow-green Chroma slide

Objective: ALL

Pixels: 1024 x 1024

Speed: 6

Zoom: 0.45

Averaging: 4 frames

Procedure:

- Open Illumination_Homogeneity_Template.czexp from LSM900 QC folder to use as template
- For each objective, set a single-channel image with each laser
- Press Start Experiment
- Save image as “field_illumination.tif” in J:\Equipment\Microscopes\Imaging Facility\[microscope]\QualityControl\yymmdd\
- Close projects

<u>Fluorescent bead imaging for PSF:</u>

Sample: 170nm green fluorescent beads

Objectives: 20x, 25x

Pixels: depends on Zoom used; adjust to pixel size

Speed: 7

Averaging: 4 lines

Procedure:

- Mount a slide with 170nm green fluorescent beads
- Open PSF_Template.czexp from LSM900 QC folder for settings
- For each objective:
    - Find FOV with beads separated and located mostly in the center.
        - Should have approximately 20 beads in FOV
    - Find out suggested XY pixel value, then triple it to oversample. Suggested sizes are:
        - 0.045µm for 20x
        - 0.080µm for 10x
    - Find out suggested z-step; halve the step size to double number of imaged planes. Suggested step sizes are:
        - 0.5µm for 20x

        - 1.2µm for 10x
    - Set a dark-to-dark z-stack. Estimated stack sizes to encompass the full PSF are:
        - 12µm for 20x
        - 40µm for 10x
    - Check that the image is not saturated in any channel
    - Image the stack
    - Name file: 25x_PSF_488
    - If the PSF appears asymmetric or otherwise indicative of a problem: redo the acquisition, with the pinhole completely open, to try to distinguish if the aberrations are due to objective or microscope.
- Save only one acquisition per lif
- Save each LIF in J:\Equipment\Microscopes\Imaging Facility\[microscope]\QualityControl\yymmdd\

<u>Check X-Y calibration of stage via stitching accuracy:</u>

Sample: Stained cells (from Abberior or other)

Objective: 25x

Pixels: “Optimal” Nyquist sampling

Speed: 7

Bidirectional scanning

Zoom: 1

Averaging: 2 lines

Procedure:

- Mount a biological sample (i.e., cells stained with 4 dyes)
- Tune settings to avoid saturation
- Set up a 3 x 3 tile scan in 4 channels
- Set 3 to 5 Z-planes
- Save LIF as “bio_sample.tif” in J:\Equipment\Microscopes\Imaging Facility\[microscope]\QualityControl\yymmdd\

When there are other objectives frequently used, but not currently mounted on the scope:

- Mount additional objectives that have not yet been tested, but are regularly used at this microscope (water dipping objectives, CLARITY objectives, etc.)
- Assign the objective to the corresponding port using the touch screen and the ID number of the objective
- For each objective, repeat the protocols for measuring the PSF with 170nm beads and for illumination homogeneity.

Shut-down:

- Find log files produced by the scanfield tests in C:/Users/Public/Pubic Documents/Carl Zeiss/ZEN/Document… and copy them to the QC folder
- Shut down the PC and microscope
- Clean objectives
- Check stage and surrounding area for spills, dust, etc.
- Replace ocular cover and dust cover on microscope

Analysis

💡 Do these steps in order, because some sections rely on previous sections (exports from ImageJ are needed for the Python code).

Excel

- Copy the laser power template into today's QC folder
- Fill it out with values from the printed template

ImageJ

- Open the 'field_illumination' file
- Use File → Save as… → Tiff for each image; one per objective.
    - They should be labeled like the following: "field_illum_05x_405_488_561_633.tif"
    - Include the objective magnification, and each wavelength used, separated by underscores.

MATLAB

- Make sure that the folder J:\Equipment\Software\MATLAB and all subfolders are listed in “Home → Set path”
- In the “current folder” window, navigate to today’s QC
- For each single-channel PSF TIF:
    - Run the following line of code (values are: filename, showPlot, zoomIn, redThreshold, showMicroscopeEvalPlot, showZProjection, beadSize, extractMetaDataFromPath)

```
psfeval('PSF_20xAir08_488.czi',1,1, 1.7, 1, 1, 0, 1);
```

    - Save the figure showing “theo, ortho, pca” and the three mean bead images as a PNG titled “25x_PSF_vals.png”
    - Save the figure with line plots as “25x_PSF.png”

Python

- Open python script “QC_plots_Zeiss.py”
    - **Modules**: laser power, detector gain linearity, detector dark noise, field homogeneity
- Change line 20 to indicate path to current QC folder, for the correct microscope and date:

```
# Input the path to QC folder for a particular date:

qc_path = r"J:\Equipment\Microscopes\Imaging
Facility\Zeiss_LSM900_O104\QualityControl\260303"
```

- Run the script.
- Check the QC folder for the following files:
    - laser_power.png
    - detector_gain_linearity.png
    - detector_darknoise.png
    - field_homogeneity.png
    - stage_repositioning.png
    - PNG for each objective for the PSFs
- Open python script “QC_over_time.py”
    - Change the variable “qc_path” to the QualityControl folder for the MICROSCOPE, not for today’s date

```
# path to microscope's QC folder
qc_path  = r"J:\\Equipment\\Microscopes\\Imaging Facility\\
Zeiss_LSM900_O104 \\QualityControl"
```

Finishing up

- Copy/paste the Quality Control Results Summary Template to today’s QC folder.

- Insert all PNGs into the correct place of the template.
- Make notes of any follow-ups that should be done at the top of the template.
- If applicable, make a folder in the “LogFiles” folder of the microscope for issues that need to be followed up on.
    - Fill out a “Template - Hardware issues” Word Doc for each issue.
    - Email the microscope company if needed.
- Record today’s date in this Excel sheet: J:\Equipment\Microscopes\Imaging Facility\ QC_Schedule.xlsx to track how often quality control is being performed.